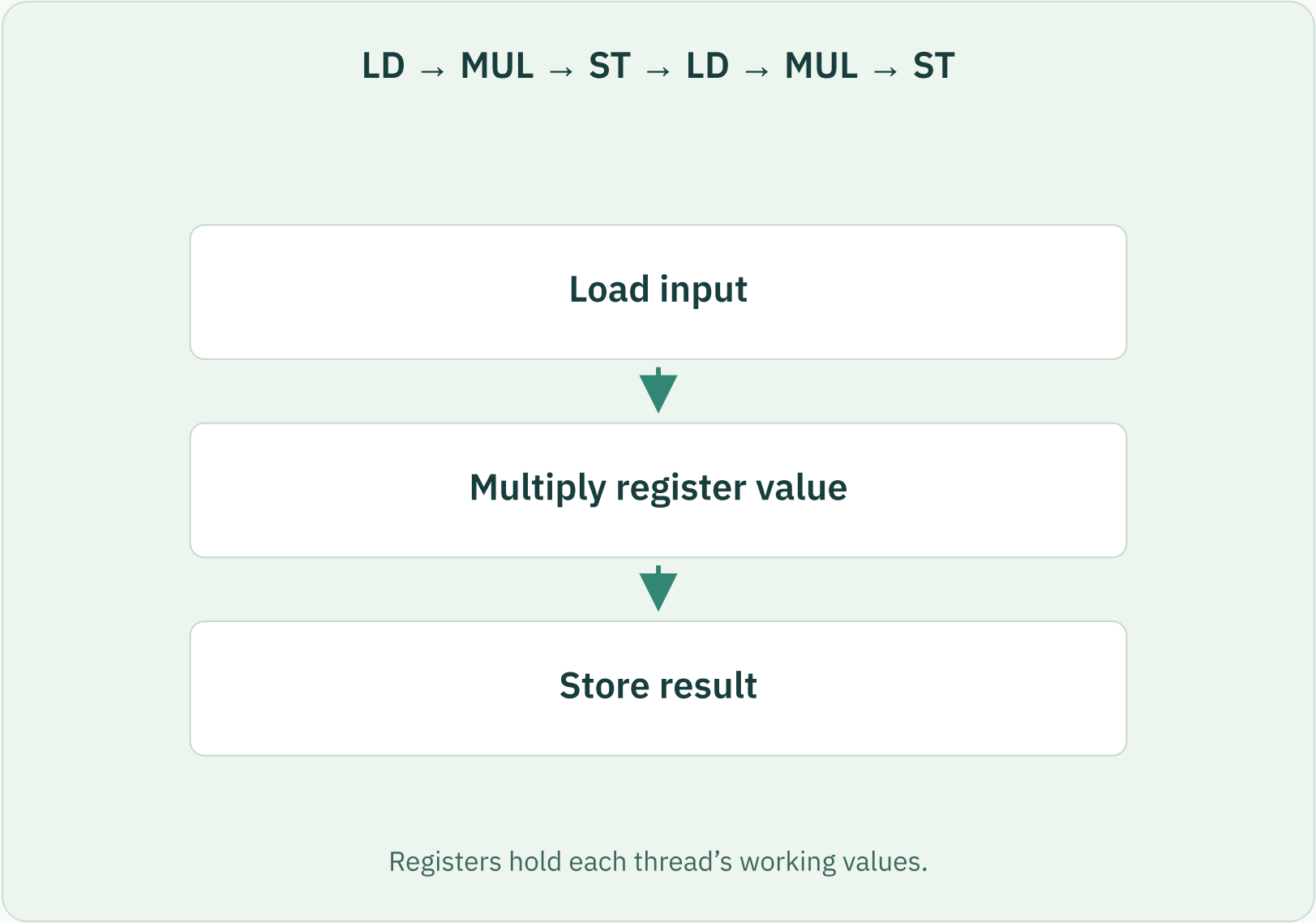
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A multiply needs its loaded input.

Load, multiply and store are separate instructions.

> Chosen instruction stream; tensor values are outside this model.



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One warp leaves waiting slots.

A load at cycle 0 becomes usable at cycle 6.

> At most one instruction starts per cycle.

LD → MUL → ST → LD → MUL → ST

1 resident warp · load latency 6 model cycles

One issue slot per cycle · W0 = warp 0

Cycle 0	Cycle 1	Cycle 2	Cycle 3	Cycle 4	Cycle 5	Cycle 6	Cycle 7
W0 LD issued	idle no issue	idle no issue	idle no issue	idle no issue	idle no issue	W0 MUL issued	W0 ST issued

State before issue at cycle 1

Warp 0: waiting

Next: MUL

Ready at cycle 6

Idle: no unfinished warp is ready

Pre-issue state · round-robin ready-warp selection

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Another ready warp can issue.

Warp zero waits while warp one starts its load.

> Resident warps share execution hardware.

LD → MUL → ST → LD → MUL → ST

2 resident warps · load latency 6 model cycles

One issue slot per cycle · W0 = warp 0

Cycle 0	Cycle 1	Cycle 2	Cycle 3	Cycle 4	Cycle 5	Cycle 6	Cycle 7
W0 LD issued	W1 LD issued	idle no issue	idle no issue	idle no issue	idle no issue	W0 MUL issued	W1 MUL issued

State before issue at cycle 1

Warp 0: waiting

Next: MUL

Ready at cycle 6

Warp 1: ready

Next: LD

Ready at cycle 0

Issue warp 1: LD

Pre-issue state · round-robin ready-warp selection

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Two waiting warps still leave gaps.

An idle slot means no unfinished warp is ready.

> Waiting and completed warps are skipped.

LD → MUL → ST → LD → MUL → ST

2 resident warps · load latency 6 model cycles

One issue slot per cycle · W0 = warp 0

Cycle 0	Cycle 1	Cycle 2	Cycle 3	Cycle 4	Cycle 5	Cycle 6	Cycle 7
W0 LD issued	W1 LD issued	idle no issue	idle no issue	idle no issue	idle no issue	W0 MUL issued	W1 MUL issued

State before issue at cycle 2

Warp 0: waiting

Next: MUL

Ready at cycle 6

Warp 1: waiting

Next: MUL

Ready at cycle 7

Idle: no unfinished warp is ready

Pre-issue state · round-robin ready-warp selection

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Readiness changes with the cycle.

Warp zero becomes ready before warp one.

> Cards show state before issue.

LD → MUL → ST → LD → MUL → ST

2 resident warps · load latency 6 model cycles

One issue slot per cycle · W0 = warp 0

Cycle 0	Cycle 1	Cycle 2	Cycle 3	Cycle 4	Cycle 5	Cycle 6	Cycle 7
W0 LD issued	W1 LD issued	idle no issue	idle no issue	idle no issue	idle no issue	W0 MUL issued	W1 MUL issued

State before issue at cycle 6

Warp 0: ready

Next: MUL

Ready at cycle 6

Warp 1: waiting

Next: MUL

Ready at cycle 7

Issue warp 0: MUL

Pre-issue state · round-robin ready-warp selection

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More warps also add instructions.

Compare idle slots without calling different workloads a speedup.

> Final store issue ends the model; no store-drain wait.

LD → MUL → ST → LD → MUL → ST

Same latency: 6 model cycles

Resident warps	Issued	Idle slots	Total cycles
1	6	10	16
2	12	8	20
4	24	4	28

Each additional warp adds six instructions.

Different total work: these times are not a speedup.

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Pause: shorten only the load wait.

Two resident warps. Load latency 3.

> Apply load start + latency, then pause.

LD → MUL → ST → LD → MUL → ST

2 resident warps · load latency 3 model cycles

One issue slot per cycle · W0 = warp 0

Cycle 0	Cycle 1	Cycle 2	Cycle 3
W0 LD issued	W1 LD issued	? predict	? predict

Warp 0 load starts: cycle 0

Warp 1 load starts: cycle 1

Cycle 2: idle or an instruction?

Cycle 3: which warp is ready?

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Issue the ready warp’s instruction.

Cycle 2 is idle; cycle 3 can start warp zero’s multiply.

> Model cycles are not elapsed GPU measurements.

LD → MUL → ST → LD → MUL → ST

2 resident warps · load latency 3 model cycles

One issue slot per cycle · W0 = warp 0

Cycle 0	Cycle 1	Cycle 2	Cycle 3	Cycle 4	Cycle 5	Cycle 6	Cycle 7
W0 LD issued	W1 LD issued	idle no issue	W0 MUL issued	W1 MUL issued	W0 ST issued	W1 ST issued	W0 LD issued

State before issue at cycle 3

Warp 0: ready

Next: MUL

Ready at cycle 3

Warp 1: waiting

Next: MUL

Ready at cycle 4

Issue warp 0: MUL

Warp 1 becomes ready at cycle 4.

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